Jiajie Zhang

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https://jiajiezhang7.github.io/

Experience

HKUST(GZ) Guangzhou, China

Research Assistant July 2026 - Present

- Work with Prof. ChanghaoChen

AI R&D Center, Central Research Institute, Wolong Electric Shanghai, China

Research Assistant July 2025 - Oct. 2025

- Supervised by Dr. Alexander Kleiner

Education

ShanghaiTech University Master of Science • 3.51/4.0 GPA

Computer Science and Technology Sep 2023 - July 2026

- Mobile Autonomous Robotic Systems(MARS) Lab
- Supervised by Prof. Soeren Schwertfeger

Zhengzhou University Bachelor of Engineering • 3.66/4.0 GPA

Automation Sep 2019 - Jun. 2023

Selected Publications

From Observation to Action: Latent Action-based Primitive Segmentation for VLA Pre-training in Industrial Settings

Accepted By CVPR 2026

Jiajie Zhang†, Sören Schwertfeger, and Alexander Kleiner

https://jiajiezhang7.github.io/latent-action-primitive-segmenter/

Generation of Indoor Open Street Maps for Robot Navigation from CAD Files

Accepted by IROS 2026

JiajieZhang†, ShenruiWu, Xu Ma and Sören Schwertfeger

https://arxiv.org/abs/2507.00552

osmAG-Nav: A Hierarchical Semantic Topometric Navigation Stack for Robust Lifelong Indoor Autonomy

Under review for Autonomous Robots

Yongqi Zhang†, **Jiajie Zhang†**, Chengqian Li, Fujing Xie and Sören Schwertfeger

https://jiajiezhang7.github.io/osmAG-Nav/

Intelligent LiDAR Navigation: Leveraging External Information and Semantic Maps with LLM as Copilot

Accepted by IROS 2025

Fujing Xie†, **Jiajie Zhang** and Sören Schwertfeger

https://arxiv.org/abs/2409.08493

Neural Surfel Reconstruction: Addressing Loop Closure Challenges in Large-Scale 3D Neural Scene Mapping
Sensors (Basel, Switzerland)
Jiadi Cui†, **Jiajie Zhang†**, Laurent Kneip and Sören Schwertfeger

https://www.mdpi.com/1424-8220/24/21/6919

WiFi-based Global Localization in Large-Scale Environments Leveraging Structural Priors from osmAG
Under review for ROBIO 2026
Xu Ma†, **Jiajie Zhang**, Fujing Xie, Sören Schwertfeger

https://arxiv.org/abs/2508.10144

Selected Projects

Bimanual Manipulation with Imitation and Vision-Language-Action Policies on Dual PiPER Arms
July 2026 – Present

- **Built an end-to-end real-world bimanual manipulation platform:** integrated dual AgileX PiPER arms, Pika teleoperation, multi-view RGB perception, ROS2, and LeRobot, covering synchronized demonstration collection, policy training, and closed-loop deployment.
- **Fine-tuned and benchmarked multiple manipulation policies,** including **ACT, Diffusion Policy, and SmolVLA**, under a unified data and evaluation pipeline to study the effects of data scale, task distribution, and policy formulation on long-horizon bimanual performance.
- **Engineered a reliable real-robot policy stack** with asynchronous inference, temporal action aggregation, dataset/runtime validation, safety gating, and structured execution traces, enabling systematic diagnosis of failures across policy, middleware, and hardware layers.

AGLoc++: WiFi-Fused Global Localization and Monte Carlo Enhanced Tracking in Hierarchical Area Graph
Nov. 2024 - May. 2026

- **Led AGLoc++ development**: Ported Area Graph LiDAR localization system to ROS2, integrated with the Nav2 stack, implemented WiFi-aided kidnap recovery, developed odometry-fused Monte Carlo tracking with advanced re-localization.
- **Enhance Robustness:** implement techniques such as clutter filtering, architectural matching, weighted ICP, corridor-aware downsampling,ensuring robust, high-precision localization in dynamic indoor environments.

https://jiajiezhang7.github.io/projects/1_agloc_project/

Campus Autonomy: Building and Navigating Autonomous Robots with Navigation2
Sept. 2024 - Jan. 2025

- Integrated advanced hardware, including the Agile X HUNTER SE **Ackermann** robot, **Hesai PandarQT64** Lidar, and **Insta360 Air** panoramic camera etc.
- Integrated navigation functionalities using the **ROS2** framework and **Navigation2** package, with testing performed in Gazebo simulator and on physical robots.
- Specialized in motion planning and control, applying and evaluating global planning algorithms (**Dijkstra, Hybrid A*, RRT***) for pathfinding within the Nav2 stack.

https://jiajiezhang7.github.io/projects/2_campus_autonomy/

Languages

English
TOEFL iBT: 106 (Reading: 30/30, Writing: 27/30, Listening: 27/30, Speaking: 22/30)